Andrew Yi Zhou

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EDUCATION

University of California San Diego **Sept. 2023 - June 2027**

B.S. in Computer Science, B.S. in Applied Mathematics

- **GPA**: 3.98
- **Coursework**: Deep Learning, Reinforcement Learning, NLP, Parallel Computing, Computer Architecture, Statistics

PUBLICATIONS

- **Andrew Y. Zhou**, Sharvaree Vadgama, Sumanth Varambally, Peter Eckmann, Michael K. Gilson, Rose Yu, "ToolMol: Evolutionary Agentic Framework for Multi-objective Drug Discovery", *ICML 2026 Workshop on Generative AI and Biology (GenBio)*, **Spotlight.** *Under Review at ICLR 2027.* arXiv:2605.12784
- Haokun Zhao, Sharvaree Vadgama, Kunhuan Liu, **Andrew Y. Zhou**, Michael K. Gilson, Rose Yu, "Synthesizability-Guided Discrete Diffusion Model for Molecular Generation", *Under Review at ICLR 2027. https://openreview.net/pdf?id=3rOXjE7u4x*
- **Andrew Y. Zhou**, Peter Eckmann, Sumanth Varambally, Michael K. Gilson, Rose Yu, "MOLLEO+: Towards Optimized Use of LLMs for Drug Discovery", *Preprint, 2025*. https://azhou2023.github.io/assets/pdf/MOLLEO+.pdf

EXPERIENCE

Spatiotemporal Machine Learning Lab, San Diego, CA **Feb. 2025 - Present**

Undergraduate Researcher

- Advised by Dr. Rose Yu and Dr. Mike Gilson in developing agents for application to small molecule drug design
- Post-trained open-source LLMs on distributed GPU clusters through Supervised Fine-Tuning, Reinforcement Learning (PPO), LoRA, distillation and RAG, improving generated molecule affinity by over 35% vs SOTA methods
- Presented first-author paper on agentic evolutionary algorithm for drug discovery at ICML GenBio & ChemAI NYC

Amazon, Santa Clara, CA **June 2026 - Sep. 2026**

Software Engineer Intern

- Developed new graph database infrastructure for storing network metadata on AWS Network State Manager team
- Built multi-threaded data stream processor for fast migration of 100M+ network objects into relational graph DB
- Optimized common network graph traversal use cases, reducing latency of relevant traversal queries by over 15x
- Engineered efficient agentic interaction with global graph DB infrastructure via contextual MCP integration

Scripps Institution of Oceanography, San Diego, CA **Nov. 2024 - June 2026**

Machine Learning Research Lead

- Leading collaboration with Scripps Institution of Oceanography to develop deep computer vision segmentation models to remotely monitor health of critical carbon-rich mangrove populations in global areas of interest
- Built UNet classifier w/ ResNet18 encoder that achieves >95% test accuracy on high resolution drone imagery
- Developed super-resolution techniques (diffusion, SRCNN, vision transformer) to upscale low-res satellite data

UCSD Center For Energy Research **June 2024 - Dec. 2024**

Research Intern

- Worked with Research Director Michael Ferry to create data visualizations of climate change in upcoming decades
- Lead developer for the Triton Energy Climate app, an interactive mobile app on App Store and Google Play that projects the effects of increasing renewable energy on CO_2 emissions; build with React Native and PostgreSQL
- Presented and shared at November 2024 United Nations COP29 Climate Change Conference with 100+ users

SKILLS & INTERESTS

- **Research Interests**: Deep Generative Modeling, LLMs, Ai4Science, Computational Drug Discovery
- **Machine learning**: PyTorch, Tensorflow, CUDA, NumPy, pandas, OpenCV, vLLM, HuggingFace, Quantization
- **Software Engineering**: React.js, Express.js, MongoDB, Node.js, MySQL, PostgreSQL, Docker, Kubernetes, AWS